Satyam Awasthi

Researcher & Software Engineer

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Professional Summary

Computer Science researcher focused on **Autonomous Systems**, **Spatial Perception**, and **Model-Based and Learning-Based Control**. My research explores how perception, predictive modeling, and planning can be integrated with control frameworks to enable robust long-horizon autonomy in robotic systems operating under uncertainty.

Research Interests

Autonomous Systems, Spatial Perception, Model-Based and Learning-Based Control, and Robot Planning, with emphasis on long-horizon decision-making under uncertainty in real-world robotic environments.

Education

University of California, Santa Barbara (UCSB) Santa Barbara, CA
M.S. in Computer Science (GPA: 3.91/4.00) Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur) Kharagpur, India
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0) Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, University of California, Santa Barbara Sep 2021 – Jun 2023
Graduate Researcher Advisor: Prof. Tobias Höllerer, Co-Advisor: Prof. Michael Beyeler

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur Jan 2016 – Dec 2018
Undergraduate Researcher

- Contributed to the development of **Eklavya 6.0**, the IIT Kharagpur autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**. `Competition Video` · `Team AGV`
- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
COMSNETS, 2022. DOI: 10.1109/COMSNETS53615.2022.9668531

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias H"ollerer.
IEEE VRW, 2024. DOI: 10.1109/VRW62533.2024.00321

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias H"ollerer.
ISMAR Adjunct, 2023. DOI: 10.1109/ISMAR-Adjunct60411.2023.00104

Research Artifacts

EyeTTS Calibration Framework 2022–2024
Calibration Framework · User Study Framework

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset 2022–2024
Dataset · User Study Videos

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Selected Projects

JitterNot Jan 2022 – Mar 2022
Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

In Motion — HAR using DeepConvLSTM Nov 2019
Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

Battleship Solitaire Solver Apr 2022 – Jun 2022
Solver-Aided Constraint Synthesis for Structured Search Problems (UCSB)

- Formulated Battleship Solitaire as a constraint-satisfaction problem in Rosette/Racket, representing grid cells and ship placements as symbolic variables subject to row/column tally, fleet-composition, adjacency, and revealed-cell constraints.
- Designed a solver-aided synthesis pipeline that parses puzzle instances, applies constraints as assertions, and uses Rosette's angelic execution to synthesize concrete board configurations satisfying all constraints.

Industry Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage **agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Teaching

University of California, Santa Barbara Santa Barbara, CA

Graduate Teaching Assistant Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024

ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.